higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

MATHEMATICS N5

(16030175)

28 March 2025 (X-paper)
09:00–12:00

Scientific calculators and drawing instruments may be used.

This question paper consists of 5 pages and a formula sheet of 4 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MATHEMATICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show all intermediate steps and simplify where possible.
5. All final answers must be rounded off to THREE decimal places (unless indicated otherwise).
6. Questions may be answered in any order, but subsections of questions must be kept together.
7. Use only a blue or black pen.
8. Sketches must be large, neat and fully labelled.
9. Write neatly and legibly.

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QUESTION 1

1.1 Determine the following limit:

$$y = \lim_{x \to 0} \frac{\tan x - x}{x - \sin x} \quad (4)$$

1.2 Determine the value(s) of x for which $f(x)$ is discontinuous if

$$f(x) = \frac{3}{e^{x^2} - e^x} \quad (2)$$

[6]

QUESTION 2

2.1 Given:

$$f(x) = \frac{4x - 3}{x + 2}$$

Determine the derivative of $f(x)$ from first principles. (4)

2.2 Determine $\frac{dy}{dx}$ in each of the following cases:
(Simplification NOT required)

2.2.1 $$y = \ln\left(\frac{2x + 1}{2x - 1}\right) + 2\tan^{-1}(2x) \quad (5)$$

2.2.2 $$y = \{\sin[\tan(\cos x^3)]\}^5 \quad (5)$$

2.3 Calculate $\frac{dy}{dx}$ if $y = (2x + 1)^{\sin 4\pi x}$ with aid of logarithmic differentiation (4)

2.4 Given the implicit function:

$$\text{cosec}^{-1} x + \cot^{-1}(x + y) = \pi$$

Determine $\frac{dy}{dx}$ (4)

[22]

QUESTION 3

3.1 Given: $f(x) = x^3 - 2x + 3$

3.1.1 Determine the coordinates of the turning points of $f(x)$. (3)

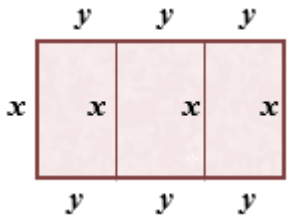
3.1.2 Draw up a table of x and $f(x)$, where x is ranging from $x = -2$ to $x = 2$; $x \in Z$. (2)

3.1.3 Draw a neat graph of $f(x)$ between these values and show the turning points on it. (2)

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3.1.4 If $x = -1{,}75$ is an approximate root of the equation $x^3 - 2x + 3 = 0$, use Taylor's /Newton's method TWICE to determine a better approximation of this root (correct root to THREE decimal figures). (4)

3.2 The owner of a dog park has **120** meters of fencing and wants to create a rectangular area that is divided into three rectangular pens, as in the diagram below. Find the dimensions of the rectangle that would yield the largest possible total area of the three pens. What is this area?



(6)

3.3 The radius of a circle is decreasing at a rate of $4\ c\text{m/s}$. How fast is the circumference of the circle changing when the radius is 8 cm.

HINT: $\mathrm{C} = 2\pi\mathrm{r}$ (3)

[20]

QUESTION 4

4.1 Determine $\int y\,dx$ in each of the following cases:

4.1.1 $y = \dfrac{\cos x}{2\sin x - 1}$ (3)

4.1.2 $y = \sin\left(\dfrac{7}{3}x\right)\sin\left(\dfrac{1}{4}x\right)$ (3)

4.1.3 $y = \dfrac{e^x}{\sqrt{3 - e^{2x}}}$ (4)

4.1.4 $y = \dfrac{2x^3 + 8x^2 - 6x + 10}{x - 2}$ (5)

4.1.5 $y = \log_2 x$ (4)

4.2 The velocity, v, of an object falling in air is given by

$$v = \int \frac{dv}{1 - (kv)^2}$$

where k is a non-zero constant.

By using partial fractions, show that

$$v = \frac{1}{2k}\ln\left(\frac{1 + kv}{1 - kv}\right) + C$$

(5)

[24]

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QUESTION 5

5.1 Determine

$$\int_1^{\infty} \frac{1}{x^3} dx \quad (4)$$

5.2 Given: $y = (x-2)^2$ and $y = -x^2 + 2x$

5.2.1 Calculate the coordinates of the points of intersection. (3)

5.2.2 Make a neat sketch to show the enclosed area, the representative strip and the point of intersection. (2)

5.2.3 Calculate the magnitude of the area in QUESTION 5.2.2 (4)

5.2.4 Calculate the volume of the solid of revolution formed when the area in QUESTION 5.2.2 is rotated about the x-axis. (5)

5.3 Calculate the second moment of area of a circular lamina with diameter 2 cm and about an axis through its centre and perpendicular to the plane of the lamina. (4)

[22]

QUESTION 6

6.1 Determine the general solution of:

$$\frac{dy}{dx} = \frac{3x(y-5)}{x^2-1} \quad (3)$$

6.2 Determine the general solution of:

$$\frac{d^2y}{dx^2} = \sin^2 3x \quad (3)$$

[6]

TOTAL: 100

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FORMULA SHEET

Any other applicable formula may also be used.

TRIGONOMETRY

$\sin^2 x + \cos^2 x = 1$

$1 + \tan^2 x = \sec^2 x$

$1 + \cot^2 x = \operatorname{cosec}^2 x$

$\sin 2A = 2 \sin A \cos A$

$\cos 2A = \cos^2 A - \sin^2 A$

$\tan 2A = \dfrac{2 \tan A}{1 - \tan^2 A}$

$\sin^2 A = {}^1\!/_2 - {}^1\!/_2 \cos 2A$

$\cos^2 A = {}^1\!/_2 + {}^1\!/_2 \cos 2A$

$\sin(A \pm B) = \sin A \cos B \pm \sin B \cos A$

$\cos(A \pm B) = \cos A \cos B \pm \sin A \sin B$

$\tan(A \pm B) = \dfrac{\tan A \pm \tan B}{1 \pm \tan A \tan B}$

$\sin A \cos B = {}^1\!/_2 [\sin(A + B) + \sin(A - B)]$

$\cos A \sin B = {}^1\!/_2 [\sin(A + B) - \sin(A - B)]$

$\cos A \cos B = {}^1\!/_2 [\cos(A + B) + \cos(A - B)]$

$\sin A \sin B = {}^1\!/_2 [\cos(A - B) - \cos(A + B)]$

$\tan x = \dfrac{\sin x}{\cos x}$; $\sin x = \dfrac{1}{cosec\ x}$; $\cos x = \dfrac{1}{\sec x}$

BINOMIAL THEOREM

$(x + h)^n = x^n + \mathrm{n}x^{n-1} h + \dfrac{n(n-1)}{2!} x^{n-2} h^2 + \cdots$

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DIFFERENTIATION

$$e = -\frac{f(a)}{f'(a)}$$

$$r = a + e$$

PRODUCT RULE

$$y = u(x) \,.\, v(x)$$

$$\frac{dy}{dx} = u \,.\, \frac{dv}{dx} + v \,.\, \frac{du}{dx}$$

$$= u \,.\, v' + v \,.\, u'$$

QUOTIENT RULE

$$y = \frac{u(x)}{v(x)}$$

$$\frac{dy}{dx} = \frac{v \,.\, \frac{du}{dx} - u \,.\, \frac{dv}{dx}}{v^2}$$

$$= \frac{v \,.\, u' - u \,.\, v'}{v^2}$$

CHAIN RULE

$$y = f\big(u(x)\big)$$

$$\frac{dy}{dx} = \frac{dy}{du} \,.\, \frac{du}{dx}$$

$f(x)$	$\frac{d}{dx} f(x)$	$\int f(x)dx$
ax^n	nax^{n-1}	$\frac{ax^{n+1}}{n+1} + c$
a	0	$ax + c$
e^x	e^x	$e^x + c$
a^x	$a^x . lna$	$\frac{a^x}{lna} + c$
$\log_e x$	$\frac{1}{x}$	___
$\log_a x$	$\frac{1}{xlna}$	___

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$\sin x$	$\cos x$	$-\cos x + c$
$\cos x$	$-\sin x$	$\sin x + c$

$f(x)$	$\frac{d}{dx}f(x)$	$\int f(x)dx$
$\tan x$	$\sec^2 x$	$ln(\sec x) + c$
$\cot x$	$-\operatorname{cosec}^2 x$	$ln(\sin x) + c$
$\sec x$	$\sec x \,.\tan x$	$ln[\sec x + \tan x] + c$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \,.\cot x$	$ln(\operatorname{cosec} x - \cot x) + c$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	——
$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}$	——
$\tan^{-1} x$	$\frac{1}{1+x^2}$	——
$\cot^{-1} x$	$\frac{-1}{1+x^2}$	——
$\sec^{-1} x$	$\frac{1}{\sqrt{x^2-1}}$	——
$\operatorname{cosec}^{-1} x$	$\frac{-1}{x\sqrt{x^2-1}}$	——
$\frac{1}{x\sqrt{x^2+a^2}}$	—	$\frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + c$
$\sqrt{a^2 - x^2}$	—	$\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + \frac{x}{2}\sqrt{a^2 - x^2} \; + c$
$\frac{1}{x^2-a^2}$	—	$\frac{1}{2a} ln\left(\frac{x-a}{x+a}\right) + c$
$\frac{1}{a^2-x^2}$	—	$\frac{1}{2a} ln\left(\frac{a+x}{a-x}\right) + c$

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INTEGRATION

$$\int f(x)\, g'(x)\, dx = f(x)\, g(x) - \int f'(x) g\,(x) dx$$

$$\int \frac{f'(x)}{f(x)} dx = ln\, f(x)\, + c$$

$$\int [f(x)]^n\, f'(x) dx = \frac{[f(x)]^{n+1}}{n\, + 1} + c$$

$$\frac{f(x)}{(ax + b)(cx\, + d)} = \frac{A}{ax\, + b} + \frac{B}{cx\, + d}$$

$$\frac{f(x)}{(x\, + a\,)^n} = \frac{A}{(x\, + a)} + \frac{B}{(x\, + a)^2} + \frac{C}{(x + a)^3} + \cdots \frac{Z}{(x\, + a)^n}$$

APPLICATIONS OF INTEGRATION

AREAS

$$A_{x\,=} \int_a^b y dx\,;\; A_x = \int_a^b (y_1 -\, y_1) dx$$

$$A_x = \int_a^b x dy\,; A_{Y=} \int_a^b (x_1 - x_2) dy$$

VOLUMES

$$V_x = \pi \int_a^b y^2\, dx\,;\; V_{X\,=\,\pi} \int_a^b (y_1^2 - y_2^2) dx$$

$$V_y = \pi \int_a^b x^2\, dy\,;\; V_{y=\,\pi} \int_a^b (x_1^2 - x_2^2) dy$$

SECOND MOMENT OF AREA

$$I_x = \int_a^b r^2\, dA\,;\; I_{y=} \int_a^b r^2\, dA$$

MOMENT OF INERTIA

$$Mass\; = density\; \times volume$$

$$M\; = pv$$

$$DEFINITION : I\; = m\, r^2$$

$$GENERAL : I\; = \int_a^b r^2 dm\; = p \int_a^b r^2 dV$$

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